

CanSat 2026 — Final Project Report

Team CAN-Team-25 · Physics Club, SVNIT

Report date: 12 September 2026 · revised at submission, 14 September 2026

Repository: github.com/KunjSorathiya/CanSat-Team-Singularity

A note on what this report claims.

*This project keeps one rule, and this report is written under it: **nothing is described as done unless there is evidence, and the evidence is named.** Owning a component is not integration. A passing test suite is not flight verification. A simulation is not a drop test.*

Where something has been measured, the measurement and its date are given. Where something is a model, it is labelled a model and the file that computes it is named. Where something has not been done, this report says so plainly rather than leaving it out — including in the sections where that is unflattering.

The CanSat and this report were submitted on 14 September 2026. Nothing in this project has flown yet — the launch is still ahead, so this report has no results section, and says so rather than implying one.

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1. Executive summary

The CanSat is a can-sized satellite that is lifted to 100 ft by a drone, released, deploys a parachute, descends at no more than 5 m/s, carries an egg chamber through landing, and streams telemetry continuously from power-on through recovery.

Where the project stands at submission, 14 September 2026:

Layer	State
Submission	The CanSat and this report were submitted (reported by the team, 2026-09-14). The launch has not happened
Software	Complete and passing 6131 automated checks on the host — flight core, telemetry protocol, ground station, web console, simulations and documentation claims
Firmware drivers	Every driver has run on real silicon: IMU, barometer, GPS, radio and microSD. The sealed flight image has been flashed and run
Electronics	Built, and every device on it works . The radio link has closed end to end. Switch, power LED and battery divider fitted; the Schottky diode never was
Structure	Printed in white PETG and assembled , egg chamber fitted, ballasted into the 450–550 g band (reported by the team, 2026-09-14)
Recovery	80 cm canopy sewn and fitted (reported by the team, 2026-09-14). Never dropped, never deployed
Flight	Nothing has flown . The launch will be the first descent, the first deployment and the first measured descent rate

The three numbers that matter most

Mass	450–550 g band	Ballasted into it before submission (reported by the team, 2026-09-14). The last scale reading on record is 280 g assembled without a parachute, 2026-09-12
Telemetry rate	1.43 Hz , then 3.11 Hz for the flight	Measured at the ground station. The rulebook floor is 1 Hz
Descent under the fitted canopy	4.37–5.00 m/s over 6.45–7.28 s across the mass band	Modelled. The cap is 5 m/s; no drop test was made

What changed at submission, and why it matters

Three days before submission the vehicle was printed, assembled and weighed at **280 g without a parachute** — and that inverted the project's largest open risk. The print came in at **≈ 128.7 g against a 193 g solid-volume upper bound, 33 % under**, so the finished vehicle would have been **105–135 g below the 450 g edge** of the rulebook's mass band.

Whether that edge binds was never answered: GEN-005 reads "*500 g (±10%)*" as a band, while GEN-006's disqualification names only *exceeding*. **The team made the question moot** by ballasting into the band, which satisfies either reading. In the same days the **80 cm canopy was sewn and fitted and the switch and power LED went in** — the last mandatory items on the build.

What was never done is a drop test. The canopy has never opened and the structure has never absorbed an arrival, so the descent rate in this report is a model, and the launch is its first test.

2. Mission and requirements

2.1 The mission

The CanSat is powered on at ground level and begins transmitting immediately. A drone lifts it to **100 ft (30.48 m)** and releases it. It deploys a parachute, descends at **no more than 5 m/s**, transmits continuously throughout, keeps transmitting for **at least five seconds after impact**, and is then recovered.

Flight is autonomous. There is no launch command, no arming button and no manual trigger anywhere in the firmware: the vehicle powers on, calibrates itself, arms itself, detects its own launch and its own landing, and keeps talking through every failure it can survive.

2.2 Requirements

Thirty requirements were extracted from the rulebook and each was given an implementation, a verification method and a status. The full traceable table is in `documentation/requirements/requirements.md`. The rulebook's own hard numbers:

Quantity	Value	Requirement
Body envelope	21 cm × 12 cm, +7 cm for the egg chamber	GEN-004
Mass	500 g ± 10 % — 450 g to 550 g	GEN-005
Release altitude	100 ft / 30.48 m , from a drone	MIS-001
Descent rate	≤ 5 m/s	REC-005
Telemetry rate	≥ 1 packet per second	TEL-004
Post-impact telemetry	≥ 5 s	REC-008
Radio	433 MHz LoRa, sync 0xF3 test / 0xA5 launch	TEL-020

Exceeding size or mass by more than 10 % is a disqualification, not a deduction. It sits on a closed list alongside unsafe deployment, no attempt at a communication system, arriving late, and code-of-conduct violations.

2.3 Two rulebook contradictions, both resolved

The original guidelines contradicted themselves on dimensions and on launch altitude. Neither was guessed at locally; both were escalated, and the **2026 revision settled both**:

Was contradictory	Now stated
Dimensions	21 cm (+7 cm for the egg chamber) × 12 cm , stated identically on pages 4 and 10
Launch altitude	100 ft, released from a drone , stated identically in both places

A third ambiguity — whether "12 cm across" meant a width or a diameter — was put to the organizers directly, because for a prismatic body the two give different answers and the difference was disqualification-class rather than a scored margin. **They confirmed on 2026-09-09 that a 12 cm sided box is acceptable.** That written confirmation must travel with the submission; it is currently recorded only in this repository.

3. Concept of operations

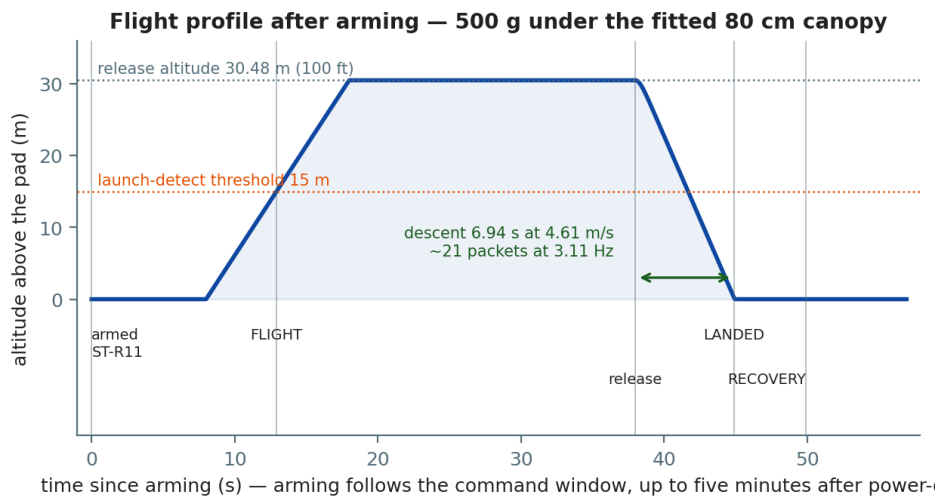


Figure 1 — the flight after arming, for a 500 g vehicle under the fitted 80 cm canopy. The submitted mass is inside the band but not recorded, so 500 g stands in for it. The descent segment is the model's closed-form solution, not height ÷ rate.

Milestone	When	Where the number comes from
Telemetry begins	Immediately at power-on, at 1.43 Hz	No trigger exists in the firmware
Command window	0 – 300 s	<code>command_window_ms</code> . The vehicle listens between packets and does not arm
Window closes	At 300 s, or on an accepted <code>MAX_RATE</code>	Whichever is first
Max rate	From the close	<code>auto_max_rate</code> — 3.11 Hz measured
Armed	3 s after the close, once recalibration settles	<code>arming_delay_ms</code> , counted from the close
Calibration times out best-effort	20 s after the close	<code>calib_timeout_ms</code>
READY → FLIGHT	15 m above the pad, held 300 ms	<code>launch_altitude_gain_m</code>
Release altitude	30.48 m	Rulebook, MIS-001
Descent duration	6.45–7.28 s across the 450–550 g band	<code>simulations/descent.py</code> , 80 cm canopy
Landing detected	3 s at rest, after an observed descent	<code>landing_confirm_ms</code>
Post-impact window	5.0 s	<code>post_impact_transmission_ms</code>

The pad phase is up to five minutes; the flight is under fifteen seconds of it.

3.1 The five-minute command window, and the one operational constraint it creates

The sealed flight image opens an uplink window for **five minutes from power-on**, during which the vehicle transmits at 1.43 Hz so that a command can be heard in the gap between packets. When the window closes — or when a `MAX_RATE` command closes it early — the vehicle goes to its max-rate pattern by itself, **discards its power-on calibration, recalibrates where it now sits**, and arms.

***The drone must not lift off until the vehicle has armed.** Launch detection is disabled for the whole of the command window, so a lift that begins inside it is never detected and the vehicle stays in READY through the entire flight. Wait for the status field to read armed — `ST-R11...` — and the station's rate to rise to about 3.1 Hz.*

Once armed the uplink is closed for the whole of flight, landing and recovery — which is every state holding a log that cannot be recreated.

3.2 The hover problem, and the guard that closed it

Writing the concept of operations found a defect that no test had: **the vehicle declared a landing while hanging under the drone, up to twelve seconds before release.**

Landing detection asked two questions — is acceleration within 2.5 m/s^2 of 1 g , and is vertical speed below 1 m/s ? **A vehicle hovering under a drone answers yes to both.** Held for three seconds, that is a landing. Run against the real state machine with a 3 m/s climb and a 20 s hover:

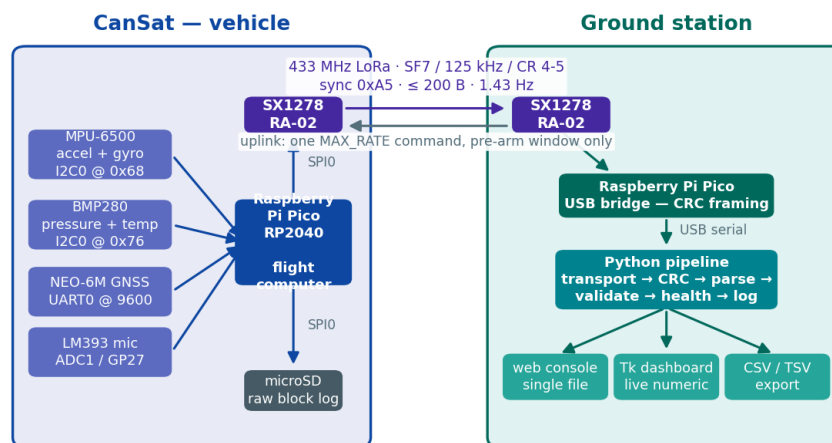
```
t= 10362 ms alt= 16.1 m rate= +3.0 m/s READY -> FLIGHT
t= 18018 ms alt= 30.0 m rate= +0.0 m/s FLIGHT -> LANDED <-- still under the drone
t= 23034 ms alt= 30.0 m rate= +0.0 m/s LANDED -> RECOVERY <-- 12 s before release
```

The existing guard did not catch it because `min_flight_ms` suppresses landing detection only for the first 3 s of `FLIGHT`, and on a drone lift `FLIGHT` is entered at 15 m during the *ascent* — so those three seconds are spent long before the hover.

The fix is a descent gate: a vehicle cannot land without descending first. The state machine latches `descent_observed_` once the vertical rate has been below -2 m/s for one second, and refuses `FLIGHT` → `LANDED` until it is set. The latch belongs to one `FLIGHT` and clears on any state change, so a descent seen earlier cannot authorise a landing later. The same reproduction now lands three seconds after touchdown.

4. System architecture

System architecture — two Picos, two identical radios, one protocol



The flight core compiles and runs on a laptop: hardware reaches it only through six abstract interfaces, which is why 5485 C++ assertions can exercise the mission logic with no Pico present.

Figure 2 — two Raspberry Pi Picos and two identical radios. The vehicle Pico runs the mission; the ground Pico is a pure bridge that frames every received payload onto USB with a CRC, so the PC can tell transport corruption apart from a malformed packet.

4.1 The dependency rule

The system is layered, and each layer depends only on the one beneath it:

```
main.cpp      watchdog, clock, configuration
|
flight::pico  hardware — mpu9250, bmp280, neo6m, sd_card, pico_radio
|
flight::      flight core — controller, state machine, scheduler, orientation,
|              calibration, faults, telemetry builder, block log
|
cansat::      shared — telemetry format and parser, SX1278 driver
|
```

```

ground::          bridge firmware and USB CRC framing
|
Python / JS      ground pipeline and interfaces

```

The flight core never includes a Pico SDK header. Hardware reaches it only through six abstract interfaces — Imu, Barometer, Gps, Radio, SdLogger, BoardIo. Tests substitute mocks; the vehicle substitutes `flight::pico::*`.

That single rule is why 5485 C++ assertions can exercise the entire mission logic on a laptop with no hardware present, and it is the reason the software was complete and tested before any component had been unpacked.

4.2 End-to-end data path

```

sensors -> controller -> plausibility gate -> bias correction -> orientation
-> AGL and vertical rate
-> TelemetryBuilder (nothing is built if any mandatory field is invalid)
-> packet string -> SX1278 -> 433 MHz LoRa -> SX1278 -> bridge Pico
-> CRC-framed USB serial
-> transport -> CRC check -> parse -> validate -> health -> log
-> dashboard, web console, CSV export

```

A packet becomes a telemetry point only if it survives every stage, and each stage's rejection is counted separately — so a transport fault is never mistaken for a sensor fault.

5. Components

5.1 Bill of materials

Subsystem	Component	Qty	Purpose	Status
Flight computer	Raspberry Pi Pico (RP2040)	2	Vehicle + ground bridge	Both built and running
Telemetry	SX1278 RA-02 433 MHz LoRa	2	Vehicle + ground radio	Link closed 2026-09-07, 66 packets, 0 % loss
Telemetry	433 MHz SMA antenna	2	Radio antennas	Mated and radiating
Telemetry	10 cm IPEX-to-SMA RG1.13	2	Radio to antenna	Fitted
Sensors	MPU-6500 (sold as MPU-9250)	1	Acceleration, angular rate	Working, but the wrong part — see below
Sensors	GY-BMP280-3.3	1	Pressure, altitude, temperature	Verified at 0x76, 83.0 Hz
Sensors	NEO-6M GPS with EEPROM	1	Position and timing	Talking, all six NMEA sentences, 0 checksum errors. No fix acquired outdoors yet
Sensors	LM393 sound module, 4-pin	1	Additional sensor — acoustic level	Fitted, logged, and transmitted as SN-
Storage	microSD card reader (3.3 V)	1	Onboard logging	Writes 100/100, sustains ~300 writes/s
Power	Orange 3.7 V 1500 mAh 25C 1S LiPo	1	Primary power	Behind the fitted switch. No Schottky — never connect USB with the battery in
Power	Manual ON/OFF switch + power LED	1 each	PWR-001, PWR-002	Fitted (reported by the team, 2026-09-14)
Structure	Cansat_D1, white PETG, 3D printed	1	Airframe and egg chamber	Printed, assembled, ballasted into the mass band
Recovery	80 cm flat canopy	1	Descent at ≤ 5 m/s	Sewn and fitted (reported by the team, 2026-09-14). Never deployed

Subsystem	Component	Qty	Purpose	Status
Prototyping	10 × 10 cm universal PCB	2	Electronics mounting	One built, one spare

Built and submitted. The only part of the design never fitted is the **Schottky diode**.

5.2 The IMU is not the part that was ordered

The module sold as an MPU-9250 answers WHO_AM_I with `0x70` — an **MPU-6500**. **Six axes, no magnetometer**. Address `0x0C` never appears on the bus, in either scan, with or without the pass-through bridge enabled. There is nothing behind it.

The consequence is that this vehicle has no absolute yaw reference. Yaw is gyro-integrated and drifts without bound; roll and pitch are unaffected, because they are referenced to gravity through the accelerometer. Every packet declares which kind of yaw it carries: YR-M for an absolute magnetic yaw, YR-G for a relative gyro integration whose zero is arbitrary. **On this vehicle it reads YR-G and always will.**

The nine-axis path — the AK8963 driver, the axis rotation, the calibration, the YR-M declaration — is implemented and tested and would work on a genuine MPU-9250. It has no source. **This is a procurement item, not a software change**, and whether a declared relative yaw is acceptable for the mandatory Ya- field is an open organizer question.

The firmware was written for exactly this case: it accepts `0x70` as a six-axis part and reports degraded attitude rather than refusing to boot, or — worse — inventing a heading from a bus that is not answering.

5.3 The additional sensor, and why a microphone

Sensor integration is capped at 25 points: 15 for the mandatory set, +5 for GPS, +5 for one more. The second +5 is taken by an **analogue microphone on GP27**, and the choice is defensible rather than opportunistic.

A microphone on a descending probe is a flight-proven atmospheric instrument. Mars 2020 *Perseverance* carried one dedicated to entry, descent and landing. The *Huygens* probe carried an acoustic sensor through Titan's atmosphere in 2005. *Venera 13* and *14* recorded wind noise on the surface of Venus. Acoustics is one of the cheapest ways to instrument a descent, which is why it keeps being flown.

What it measures here, in order of confidence:

Use	Confidence	Why
Landing detection	High	Impact is an impulsive transient tens of dB above anything else in the flight
Flow noise against descent rate	Good	Unshielded electrets are extremely sensitive to airflow. On a descending body that is normally a defect; here it is the measurement
Canopy oscillation	Partial	Canopy modes are 0.5–3 Hz. The level is computed every loop tick but recorded once per packet — 3.11 Hz in flight — so only modes below about 1.5 Hz can be resolved; faster ones alias
Deployment transient	Moderate	The crack is sharp and loud, but it happens exactly when flow noise is highest
Absolute sound pressure level	None	No calibration, no reference, and an unrecorded gain trimpot
Frequency spectra	None	A 0.5 ms window resolves ~2 kHz upward, which is not where the content is

Two channels are recorded, deliberately. `sound_mv_pp` is *how loud*; `sound_gate_pct` is *what fraction of the window was loud*. A sharp crack and a steady roar can reach the same peak and mean opposite things, and the duty cycle is what separates them.

What it does not measure, stated before a judge asks: the level is a relative peak-to-peak envelope in millivolts, **not a sound pressure level**. Reporting decibels would need a calibrated reference source and a record of the module's gain trimpot position, and this project has neither. Values are comparable within one flight at one gain setting and with nothing else. That limitation is written into the firmware comments, the requirements row, and the log column name.

6. Electrical design

6.1 The power question, answered by measurement

Three hardware blockers held the design up for days. All three are closed, and how they closed is more useful than the fact that they did.

No external regulator is needed. An AMS1117-3.3 was assessed and rejected — a full cell at ≈ 4.2 V does not clear its high-load dropout. The Pico's own regulator was then measured carrying every load on the vehicle:

Condition	Rail voltage
45 back-to-back radio transmits	3.28–3.29 V
100 % microSD write duty	3.28–3.30 V

One rail, no external part. The microSD reader turned out to be a **3.3 V board** with no regulator and no level shifter — its supply pin is printed 3V3 and its entire parts list is four 10 k Ω pull-ups and two capacitors. The listing that described a 4.5–5.5 V board described a different product.

6.2 The power path

```

LiPo --[ ON/OFF switch ]--[ Schottky ]--> Pico VSYS --> Pico 3V3(OUT) --> everything
      |                                     |
      +--[ 1 kOhm ]--> power LED          +-- MPU-6500, BMP280 (I2C0)
                                           +-- SX1278 RA-02   (SPI0)
                                           +-- microSD reader (SPI0)
                                           +-- NEO-6M       (UART0)
                                           +-- LM393 microphone (ADC)

```

The power LED hangs off the switched rail through a series resistor, not off a GPIO. PWR-003 requires it to light *immediately* on power-up, and a firmware-driven LED would wait for boot. The firmware separately drives a status LED on GP14 whose blink rate encodes the mission state.

The switch, the power LED and the battery divider were fitted before submission. The Schottky diode was not — about ₹10, and the one part of this drawing that does not exist on the vehicle. Without it USB back-powers the LiPo, so **a USB cable must never be connected while the battery is in**, switch closed or not.

6.3 Bus discipline

Two shared buses, and a lesson from each:

- I2C0 (GP4/GP5)** carries the IMU at 0x68 and the barometer at 0x76. Both verified together on the soldered board; 0x0C correctly absent.
- SPI0 (GP16/GP18/GP19)** carries the radio and the microSD, with separate chip selects on GP17 and GP6. **This bus produced the single most instructive defect in the project** — see F-16 in section 16.

The netlist at `electrical/schematics/vehicle-netlist.tsv` is **generated from BoardPins in the firmware**, and the generator refuses to run if the two disagree. The wiring diagrams are generated from the same source. A pin cannot drift between the code and the documentation.

7. Mechanical design and the as-built vehicle

7.1 The design

Cansat_D1 is modelled in Fusion 360 and exported to STEP. Every dimension quoted in this project is **read out of the STEP file by tools/cad_dimensions.py**, and the build fails if the documentation and the model disagree.

Solids	one, Body1, 56 faces
Bounding box	118.5 × 115.0 × 110.0 mm
Cross-section diagonal	159.1 mm
Height against the 210 mm allowance	91.5 mm unused
Clearance per side against the 120 mm section	2.5 and 5.0 mm

It is an **open box frame**: two solid side panels, two faces opened out with large arched cutouts, a central vertical spine, harness slots top and bottom, and a rectangular cutout for the switch with two LED holes beside it.

The board fits flat. A 100 × 100 mm perfboard sits inside a 115 × 110 mm section with 15 and 10 mm to spare for walls and standoffs. An earlier version of the design reasoning said it could not and should be mounted edge-on as a spine; that was reasoned against a 120 mm *circular* section, before there was a design, and it is withdrawn.

7.2 Material — white PETG

The rulebook permits any outer material and offers a **bonus for sustainable or unconventional materials**. PETG was chosen and the reasoning recorded, because section D expects the reasoning as much as the choice:

- **Tougher than PLA and far easier than ABS.** PETG does not go brittle the way PLA does, which matters for a part whose job is to survive exactly one impact.
- **Prints without an enclosure**, with little warping. ABS's toughness comes with warping and fumes a school workshop usually cannot manage.
- **Impact energy goes into deformation rather than fracture** — a bent frame is a recovered vehicle, a shattered one is not.

Print orientation was decided before printing and recorded: as modelled, sitting on its base. It was the single free variable that changes the part's strength, it costs nothing at slicing time, and it cannot be changed afterwards. Layers stack vertically, so the weak directions are tension normal to the layers and interlayer shear.

White was a finish decision: it shows a clean print rather than hiding a poor one, it is the easiest colour to photograph against any background for the mandatory views, and it runs cooler in sunlight on a pad than a dark part would.

7.3 The mass budget

Mass budget — from the last scale reading to the submitted vehicle

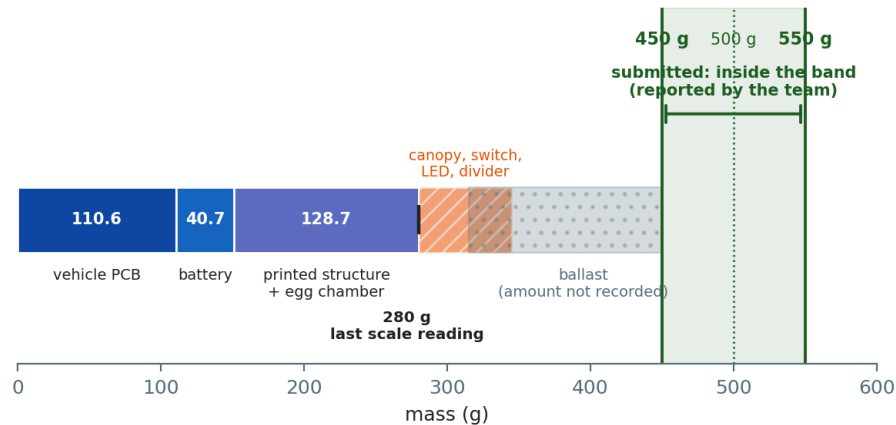


Figure 3 — from the last scale reading to the submitted vehicle. Everything to the right of 280 g went in after that reading, and the submitted total is reported, not weighed on record.

Item	Mass	How
Assembled vehicle PCB, no battery	110.573 g	Weighed 2026-09-09
Battery — Orange 1500 mAh 1S LiPo	40.726 g	Weighed 2026-09-09
Electronics, all-up	151.299 g	Sum
Printed structure + egg chamber	≈ 128.7 g	By difference
As-built vehicle, no parachute	280 g	Weighed 2026-09-12 — the last scale reading on record
Parachute, lines, harness	30–55 g	Fitted after that reading
Switch, power LED, divider	5–10 g	Fitted after that reading
Ballast	to the band	Added before submission; amount not recorded
Submitted vehicle	450–550 g	Reported by the team, 2026-09-14

What was measured and what was derived. The scale figure is the **280 g assembled vehicle**. The electronics inside it were weighed separately three days earlier. The structure line is therefore arithmetic on two measurements, not a third one: $280 - 151.299 = 128.701$ g, quoted as ≈ 128.7 g because the 280 g came off a scale reading whole grams. **The submitted mass is a third kind of number** — a report, not a measurement — and the table says so.

7.4 The estimate was 33 % high, and that is the interesting part

Predicted, solid volume $\times 1.27$ g/cm ³	193.0 g
Printed — and that includes an egg chamber the estimate did not	≈ 128.7 g
Error	–64 g, –33 %

Infill is the whole of it. The 193 g assumed no infill saving at all; a real slice of a thin-walled open frame is mostly perimeter and air. The estimate was labelled an upper bound and behaved like one — but the size of the gap is what made the vehicle light.

7.5 The risk, and how it closed

At 280 g, the finished vehicle projected to **315–345 g — 105 to 135 g under the 450 g edge** of the band. Two routes were on the table:

- **Re-print heavier.** The printed part had ~ 64 g of headroom to its 193 g solid volume, and higher infill adds material exactly where a printed part is weakest. Not enough on its own.

- **Ballast.** Cruder, and it earns nothing in section D — but it closes any gap.

The team ballasted into the band before submission. That also made the open question — whether 450 g binds at all — stop mattering, because a vehicle inside 450–550 g satisfies both readings of the rule. **The final mass was not recorded**, and weighing the vehicle before the launch is worth doing: it decides which modelled descent the flight is compared against.

7.6 What was never done mechanically

- **A drop test.** The only measurement of the drag coefficient the descent model rests on, and the only check that the printed structure survives an arrival. The launch is the first.
- **Calipers on the printed envelope.** Every dimension held here is read from the STEP, and a printed part is not its model. 2.5 mm per side is the whole clearance.
- **Photographs of the built vehicle** in this repository. Mandatory media, and section D's 15 aesthetics-and-build-quality points are judged from those images.

8. Structural simulation

Three static-stress studies were run in Fusion 360 on `Cansat_D1`.

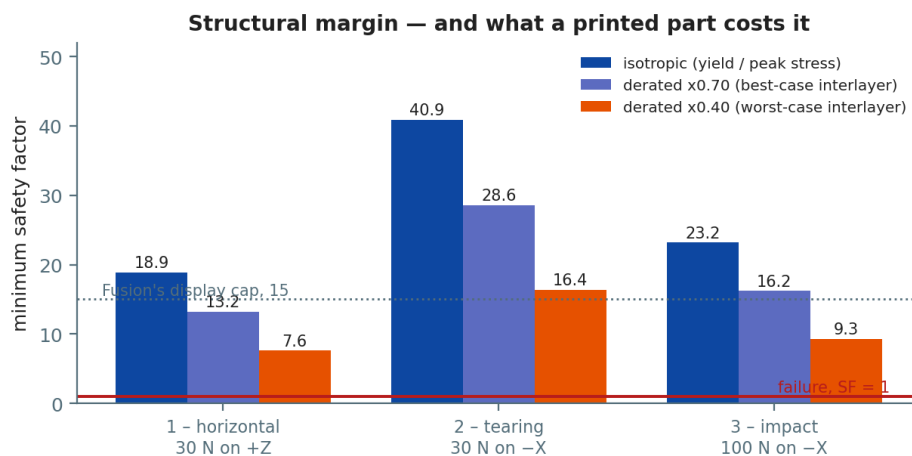


Figure 4 — safety factor per load case, isotropic and derated for the anisotropy of a printed part.

Study	Load	Mesh	Max von Mises	Yield ÷ stress
1 · Horizontal	30 N on +Z	5838 nodes	2.885 MPa	18.9
2 · Tearing	30 N on -X	5838 nodes	1.330 MPa	40.9
3 · Impact	100 N on -X	7148 nodes	2.345 MPa	23.2

The structure is nowhere near failing in any of the three. Four caveats, none of them small, and all of them stated because the numbers are useless without them.

8.1 The report says the design breaks. It does not.

All three exports contain the sentence *"the design is expected to bend permanently or break."* **It is template text.** The sentence sits above a block containing *both* the "Below Safety Factor Target" and "Above Safety Factor Limit" advice lists — which is what a template emits when it renders every branch rather than the one that applied.

And the reported minimum of 15 is a display cap, not a result. All three studies report exactly 15 while their peak stresses differ by more than a factor of two, and three different load cases cannot all minimise at exactly the same

value. Fusion's safety-factor legend caps at 15 by default, so "15" means *at or above the cap, everywhere on the part* — which is consistent with the yield-derived figures above and is the stronger statement of the two.

8.2 The simulation material is PET, not PETG

Property	In the study (PET)	Typical bulk PETG
Density	1.541 g/cm ³	~1.27 g/cm ³
Young's modulus	2757.9 MPa	~2000–2100 MPa
Yield strength	54.40 MPa	~50 MPa

PET is stiffer, denser and slightly stronger, so the studies are **mildly optimistic on strength and materially wrong on mass**. The mass error turned out larger than density alone: the printed part weighs ≈ 128.7 g where the PET density implies 234.2 g and the $\times 0.82$ density correction implies 192.0 g. The remainder is infill, which the studies do not model at all.

8.3 A printed part is not isotropic, and this is the bigger caveat

Inter-layer adhesion in Z is typically **40–70 % of in-plane strength**. With the orientation decided — printed on its base, layers stacking vertically — the derating is arithmetic:

Study	Isotropic	$\times 0.70$	$\times 0.40$
Horizontal	18.9	13.2	7.6
Tearing	40.9	28.6	16.4
Impact	23.2	16.2	9.3

Even at the pessimistic 40 %, and even taking the capped 15 rather than the yield-derived figure, the effective safety factor is 6. That is a structure with margin, not one being argued into compliance.

8.4 The 100 N impact load is an assumption — and the mass band brackets it

100 N came from a 500 g vehicle at 5 m/s arrested in 25 ms. The submitted vehicle is somewhere in the 450–550 g band, and under the fitted canopy its landing momentum brackets the studied case:

Case	Mass	Rate	Momentum	Force at 25 ms	at 10 ms
Study 3's assumption	500 g	5.00 m/s	2.50 N·s	100 N	250 N
Bottom of the band, ISA	450 g	4.37 m/s	1.97 N·s	79 N	197 N
Nominal, ISA	500 g	4.61 m/s	2.31 N·s	92 N	231 N
Top of the band, 35 °C	550 g	5.00 m/s	2.75 N·s	110 N	275 N

So the studied load sits inside the band, not above it. At the very top — 550 g on a hot day — the landing is 10 % harder than study 3, which scales its capped safety factor of 15 to about 13.6 and its worst derated figure to about 8. The margin absorbs it easily, but the earlier claim that the as-built mass made the study conservative no longer holds once the ballast went in, and this table replaces it.

It does not retire the caveat. All three studies load *horizontally*, and a vehicle under a canopy lands **base-first** — **along the build axis, which is the print's weak direction**. Lower force in the wrong direction is still the wrong direction. The vertical impact case is the one load case never run, and the launch will answer it first.

9. Descent simulation and the parachute

`simulations/descent.py` sizes the canopy and models the fall. It runs in the host test suite, so a change that breaks the physics fails the same build as a change that breaks the firmware, and it is pinned to closed-form limits that can be checked by hand.

9.1 What it computes

- Canopy area and flat diameter for a target descent rate, from $S = 2mg / (\rho \cdot C_d \cdot v^2)$.
- The descent rate a canopy you already have will produce.
- Descent **time**, from the closed-form solution of $m \cdot dv/dt = mg - \frac{1}{2} \rho C_d S v^2$ rather than height ÷ rate — the vehicle starts at rest and accelerates into terminal, and over a 30 m drop that transient is a third of a second.
- The number of telemetry packets the descent yields.
- Air density from the gas law, using **pressure and temperature the vehicle itself measures**. A 35 °C launch day is ~6.5 % thinner than ISA, which is ~7 % more canopy.

What it does not compute: canopy opening dynamics, oscillation, the drag of the bare vehicle before deployment, or wind drift. Deployment delay is modelled as free fall, which is pessimistic on altitude and therefore safe.

9.2 The canopy, and why it is sized at 550 g

Case	Flat diameter	Rate	Time
500 g, ISA 15 °C, vented flat circular	73.7 cm	5.00 m/s	6.45 s
550 g, 35 °C — size to this	80.0 cm	5.00 m/s	6.45 s

Size at the top of the mass tolerance, not at the nominal mass. Canopy area is linear in mass, so $\pm 10\%$ of mass is $\pm 10\%$ of area but only ~5 % of diameter. A canopy sized at 500 g and flown at 550 g **breaks the 5 m/s cap**; one sized at 550 g is compliant across the whole band and costs 6 cm of cloth. A test asserts exactly this.

9.3 The fitted canopy across the mass band

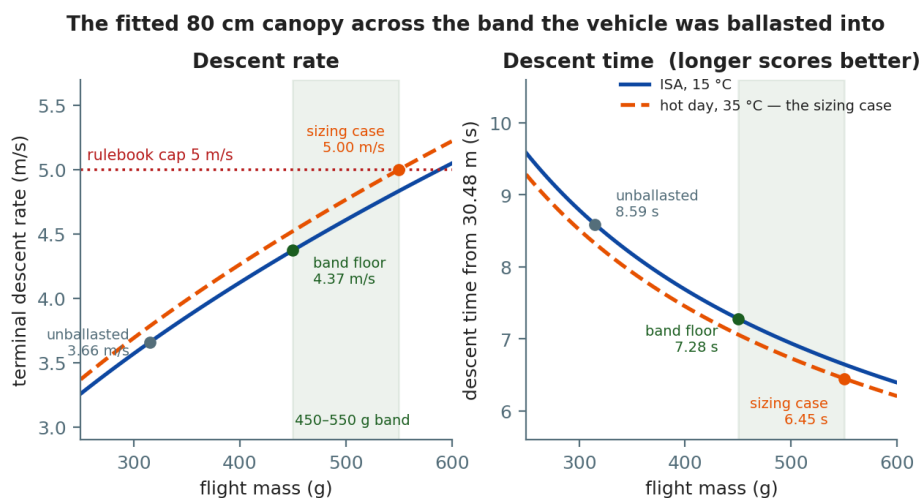


Figure 5 — the 80 cm canopy across the 450–550 g band the vehicle was ballasted into, on an ISA day and on the 35 °C day it was sized against. The unballasted 315 g point is kept for comparison.

Flight mass	Rate	Descent time	Packets at 3.11 Hz
450 g, ISA — the bottom of the band	4.37 m/s	7.28 s	~22
500 g, ISA — nominal	4.61 m/s	6.94 s	~21

Flight mass	Rate	Descent time	Packets at 3.11 Hz
550 g, 35 °C — the sizing case	5.00 m/s	6.45 s	~20
315 g — unballasted, for comparison	3.66 m/s	8.59 s	—

Every row in the band clears the 5 m/s cap, which is exactly what sizing at the top of the tolerance was for. Two things follow:

- **The ballast cost descent time.** Unballasted, the vehicle would have fallen for 8.59 s; in the band it falls for 6.45–7.28 s. Section C scores descent time comparatively, so a vehicle ballasted to the bottom of the band keeps more of that than one ballasted to the middle — which is one more reason to record the submitted mass.
- **About twenty packets of descent go over the air**, one in three carrying GPS and sound, because the vehicle flies at max rate after its command window.

9.4 Three constraints on the canopy that are not the diameter

- It **must not be tightly packed** — external or semi-exposed, so it deploys immediately on release (REC-003, REC-004). A chute stuffed inside the body risks the 5 deployment points *and* a disqualification for unsafe deployment.
- Descent must be **stable, without tumbling or spinning** (REC-006), scored comparatively. An unvented flat circular canopy oscillates. A central vent of roughly 10 % of the diameter costs a little drag and buys a great deal of stability; a cruciform is better still and is easy to sew.
- The structure must be **intact after landing** (REC-007) and telemetry must continue for at least 5 s (REC-008) — which means the antenna and the battery connection have to survive the arrival.

9.5 The uncertainty paper cannot close

Drag coefficient is the dominant term, and the spread between canopy types is larger than every other term in the model combined:

Canopy	Cd	Diameter at 500 g
Cruciform	0.85	69.3 cm
Flat circular	0.80	71.4 cm
Vented flat	0.75	73.7 cm
Hemispherical	1.40	54.0 cm

Only a drop test closes it: a known mass, a known height, a stopwatch, and the measured rate fed back into the model.

10. Flight software

The vehicle runs a **single non-blocking loop on a 2 ms tick**. Nothing waits on hardware, nothing allocates, and every recovery path is bounded.

10.1 The flight loop

```
poll(now_ms)
| -- capture epoch on first call; mission_ms = now_ms - epoch_ms
| -- gps.poll()          bounded UART drain — never blocks on a fix
| -- every 33 ms         acquire_sensors()
| -- every tick          run_calibration()
| -- every tick          feed_state_machine()
| -- every 700 ms        emit_telemetry()
| -- every 2000 ms       logger.flush()
| -- every 1000 ms       sample_battery()
```

```

|-- every 1000 ms    refresh_health()
+-- every tick      update_led()      blink rate encodes mission state

```

The ordering is deliberate. GPS is drained first so a full UART buffer can never back up — at 9600 baud the RP2040's 32-byte FIFO fills in 33 ms, and `validate_config()` enforces that the tick is short enough. Calibration and the state machine run *before* telemetry, so every packet carries the state that matches the samples inside it.

`PeriodicTask` self-corrects after a stall: if the loop falls behind by more than one period, the next due time re-anchors to now + period rather than firing a catch-up burst.

10.2 Sensor acquisition

```

acquire_sensors()
imu.read()
| -- not valid or not finite? -----> stale check: 2000 ms without a good read
|                                     raises imu_stale + orientation_invalid
| -- outside datasheet bounds? -----> reject the sample, DROP the previous value,
|   (accel > 170 m/s², gyro > 2200 dps)   raise sensor_implausible
+-- good -----> feed the calibrator with the RAW sample,
                                     subtract bias, orientation.update(dt)

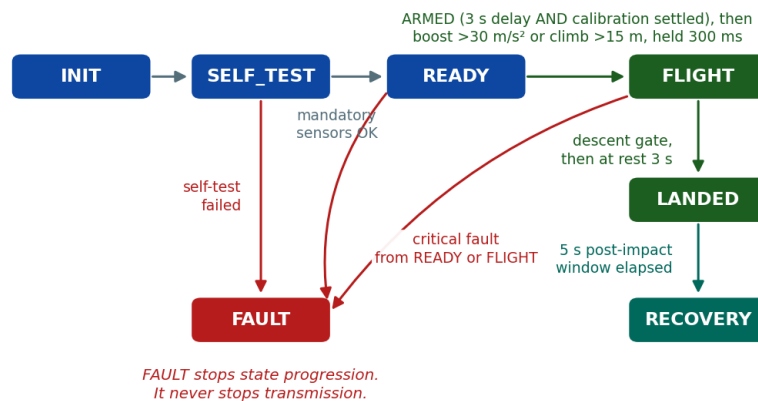
baro.read()
| -- outside 30-115 kPa or -50..95 C? ----> stale check, baro_stale after 2000 ms
+-- good -----> AGL = altitude - ground baseline
                                     vertical rate EWMA (0.7 old, 0.3 new)

gps.latest() + last_fix_ms
+-- a fix is used only while it is being renewed

```

10.3 The mission state machine

Mission state machine — every transition and the guard on it



Three guards, and each exists because something got past the obvious test:

- `READY` → `FLIGHT` refuses to fire until the vehicle is `ARMED`, so a startup glitch cannot launch it.
- `FLIGHT` → `LANDED` refuses to fire until a real descent has been seen. A drone hover answers every other landing test, and used to declare a landing 12 s before release (F-20).
- `RECOVERY` is terminal. The machine never returns to `FLIGHT`.

Figure 6 — every transition and the guard on it.

10.4 Six behaviours worth knowing about

It calibrates itself on the pad — and never lets that block the mission. While stationary it collects IMU and barometer samples and captures gyro bias, accelerometer offset and the barometric ground reference, so altitude reads ≈ 0 on the pad rather than the ~ 23 m an uncalibrated barometer showed on the bench. Acceptance needs 80 samples with per-axis gyro standard deviation under $2^\circ/\text{s}$ and acceleration magnitude within 1.5 m/s^2 of 1 g . If the vehicle is not still, calibration retries until a 20 s timeout and then resolves best-effort: **the barometric reference is still used, but gyro and accelerometer bias are not applied** — a bias measured while moving would be worse than no correction. A warning is raised and the mission continues.

A startup glitch cannot trigger a false launch. READY → FLIGHT is refused until the vehicle is armed: the 3 s arming delay has elapsed *and* calibration has settled. Then the launch condition — acceleration above 30 m/s² or a climb of more than 15 m — must hold continuously for 300 ms.

An implausible reading is treated as worse than no reading. A value outside datasheet-derived bounds is not merely skipped — **the previously held value is dropped too**, so the vehicle never coasts on data from a sensor that is actively wrong. Staleness is time-based: 2 s without a good read raises the fault, so one failed read changes nothing.

Wrong data is never transmitted, and suppression never fakes packet loss. If any mandatory field cannot be trusted, no packet is produced — **and the packet number is not consumed**. Transmitted packets stay strictly sequential, so a gap at the ground station means radio loss and nothing else.

Every peripheral can fail without stopping the mission. GPS missing? The optional fields are omitted. SD failing? Logging disables itself after 10 consecutive write failures. Radio down? Bounded re-initialisation with a 1 s back-off, never a blocking retry loop. Only a total loss of mandatory sensing counts as critical — and even then **telemetry keeps running in FAULT**.

It survives its own crashes. A 2 s hardware watchdog reboots a hung loop; telemetry restarts automatically and the reboot is reported as a fault so the ground station can see it.

10.5 The fault model

Faults are **classified, not fatal**:

Class	Effect	Examples
Warning	Counted and reported; nothing else changes	Calibration timed out, low battery, GPS stale
Degraded	A field or a subsystem is dropped; telemetry continues	SD logging disabled, radio re-initialising, sound stale
Critical	→ FAULT, state progression stops — transmission does not	Bad configuration, failed self-test, both IMU <i>and</i> barometer stale

The design rule behind the whole table: **telemetry never stops**. No peripheral failure and no state, FAULT included, suppresses telemetry that can still be produced correctly.

10.6 Design rules

These are the invariants the code is built around. Breaking one is a design change, not a refactor.

1. **Telemetry never stops.**
2. **Wrong data is worse than no data.** Mandatory fields that cannot be trusted suppress the packet rather than transmitting a plausible-looking wrong value.
3. **Sequential numbering is over transmitted packets.** Suppression does not consume a number.
4. **The flight core knows no hardware.**
5. **Nothing in the flight loop blocks or allocates.** Fixed-size buffers, bounded loops.
6. **Every constant says where it came from** — rulebook, datasheet, textbook or engineering choice, named in a comment at the definition.
7. **Predictions stay labelled as predictions** until a measurement replaces them.

10.7 Code originality

Every driver in the flight path is written here, against its datasheet and register map: **MPU-9250 family, BMP280, NEO-6M, SX1278 and the SD card protocol**. There are no third-party libraries anywhere in the flight path. The BMP280 compensation reproduces the datasheet's own reference vector; the SD card driver implements CMD0/CMD8/ACMD41 initialisation and SDHC vs SDSC block addressing; the FAT32 reader walks an MBR, a BPB and a cluster chain.

11. Telemetry protocol and the radio link

11.1 The packet

CAN-Team-25; P-042; Ti-00:01:23:450; A-18.4; Pr-100821.33; T-24.6; Ro-2.1; Pi--1.4;
Ya-15.9; AX-0.12; AY--0.31; AZ-9.79; GP-Lat-21.16450; GP-Lon-72.78480;
GP-Alt-21; SN-412.5; ST-F110;

Field	Meaning	Format
CAN-Team-XX	Team identifier	Correct identifier in every packet
P-XXX	Packet number	Starts at P-001, increments sequentially
Ti- HH:MM:SS:MS	Mission clock since power-on	
A-XXX.X	Altitude	Metres, 1 decimal
Pr-XXXX.XX	Pressure	Pa, 2 decimals
T-XX.X	Temperature	°C, 1 decimal
Ro- / Pi- / Ya-	Roll / pitch / yaw	Degrees, 1 decimal
AX- / AY- / AZ-	Acceleration	m/s ² , 2 decimals
GP-Lat / GP-Lon / GP-Alt	GPS position, only when a fix exists	Optional
SN-	Acoustic level, mV peak-to-peak	Optional
ST-	Status: state letter, armed, calibrated, active faults — ST-F110 is FLIGHT, armed, calibrated, no faults	Optional; dropped from any packet it would push past 200 bytes

Precision is enforced exactly, in the formatter and in all three parsers. The C++, Python and JavaScript implementations are held to **one shared fixture file**, `test-data/protocol-fixtures.tsv`, so they cannot drift apart.

11.2 Two packet shapes, and a 200-byte ceiling

Shape	Carries	Worst case
Rich	The twelve mandatory fields, then GP-Lat, GP-Lon, GP-Alt, then SN-	209 B , held to 200 B on the air
Lean	The twelve mandatory fields only	147 B

The 200-byte ceiling is not ours. The organizers' receiver — an ESP32 running the arduino-LoRa library, shared with the team on 2026-09-11 — reads into a 201-byte buffer and **discards anything over 200 bytes**, printing an error and scoring nothing. The competition scores what *their* station receives, so 200 is the ceiling whatever this project's own bridge can hear.

At 6 decimals of latitude and 1 of altitude the GPS block was 55 bytes and mandatory + GPS was 202 — two bytes over. Printed to what the NEO-6M actually resolves — **5 decimals (1.1 m, against its ~2.5 m horizontal error) and whole metres** — it is 51, and the total is 198. The SD log keeps the full precision.

The byte figures are defended by construction, not by arithmetic. A test builds the widest packet each shape can produce — packet number 4294967295, a 99:59:59:999 clock, every axis at full scale — and checks it against its constant. **That test corrected this design twice**, once by 2 bytes on the lean packet, which crossed a LoRa symbol boundary and cost 5 ms of airtime on every packet.

11.3 The radio

Parameter	Value	Fixed by
Frequency	433 MHz	Rulebook band
Spreading factor	SF7	Engineering choice — airtime
Bandwidth	125 kHz	Engineering choice
Coding rate	4/5	Engineering choice
TX power	17 dBm (PA_BOOST)	RA-02 maximum without PA_DAC
Sync word	0xA5	Rulebook — and the organizers' station listens on nothing else

SF7 is airtime-driven, not range-driven. A ~190-byte packet needs 943 ms at SF9/125 kHz and about 302 ms at SF7/125 kHz; only the latter meets 1 Hz with margin. The link budget still closes with tens of dB to spare at the mission's 30 m range.

Both Picos fly 0xA5, for testing as well as for the launch. On the 2026-09-10 range test the vehicle was on the test word 0xF3, and the organizers' station heard only the few packets an SX127x sync filter lets through while the team's own station heard every one. That is a silent, total loss of telemetry that looks exactly like a dead radio, and it is the reason both images now fly the official word all the time.

One definition of the modem parameters is shared by both ends — `link_profile.hpp` — because before it existed the two carried separate copies and agreed only by coincidence.

11.4 The rate

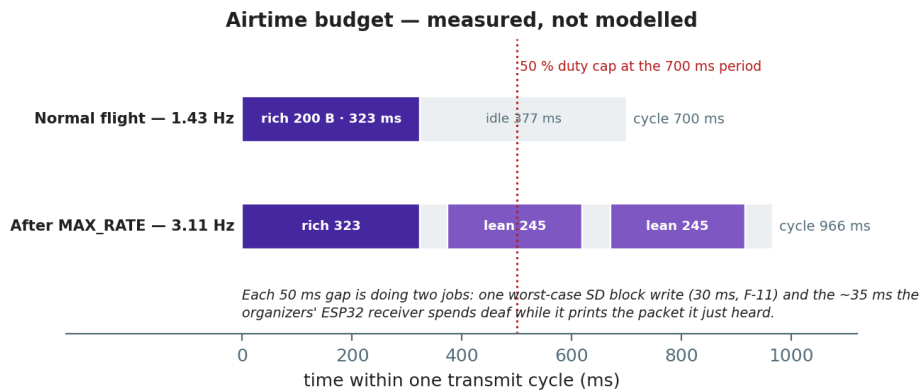


Figure 7 — where the transmit cycle goes, sized from measured airtime.

	Command window	Flight — after the window closes
Period / cycle	700 ms	966 ms (374 + 296 + 296)
Packet rate	1.43 Hz	3.11 Hz measured
Sensors on the air	1.43 Hz	1.04 Hz
Duty	46.2 % measured	84 %

The rulebook's 1 Hz is a floor this vehicle cannot be configured onto. Three independent mechanisms enforce it: a `static_assert` refuses to compile a profile above 950 ms, `validate_config()` refuses to run one, and `check_doc_claims.py` refuses to pass a repository whose documentation disagrees. A flight build silently running at 1 Hz is exactly the failure that is designed against.

Every slot is measured airtime + 50 ms, rounded up — not modelled airtime. The model reads 1.8 % low against this hardware, reproduced to 0.1 ms across four sessions.

The 50 ms guard does two jobs in the same gap. One is the vehicle's own worst-case SD block write — 30 ms, seen on two boards in two of five sessions. The other is the organizers' receiver, which is **deaf for about 35 ms after every packet** while it prints what it just heard at 115200 baud. The previous 40 ms guard covered the card and not that.

12. Onboard logging

The onboard log writes into **raw 512-byte blocks with no filesystem**, rewriting its header after every record. A brownout or an impact reset therefore resumes at the correct block instead of overwriting flight data.

The log writes one row per telemetry packet, appended in `Controller::emit_telemetry()` as the packet goes on the air — so it runs at the radio's cadence, **3.11 Hz in flight**, not at the 30 Hz sensor rate. *An earlier version of this report said 30 Hz; the code says otherwise, and the 36.8-minute bench log confirms it: 2209 rows at the 1 Hz period it then carried.* Each row carries satellite count, HDOP and full-precision position that never go on the air.

The SD log is still the primary record, but for completeness rather than density. It holds the same ~20 descent rows the radio sends, with none lost to the link and with every column the packet leaves out. Every descent-rate number in the post-flight analysis will come from the card.

Two supporting tools exist: `tools/prepare_sd_card.py` creates and sizes the log file, and `tools/inspect_sd_log.py` reports its extent layout. The firmware maps log block numbers through an extent list of up to 16 runs, so **a file in a few pieces is not a fault** — only one broken into more runs than that is refused.

13. Ground station

Three interfaces over one pipeline:

Interface	Use
Web console — ground-station/web/index.html	Single file, no build, no dependencies. Demo replay, file replay, or live Web Serial
Tk dashboard — <code>python src/main.py live --port COM5 --framed</code>	Live numeric view, with plots when matplotlib is installed
CLI replay — <code>python src/main.py replay ... --export out.csv</code>	Offline parse, validate, log and export

```
transport (serial / file / loopback)
-> CRC-16 frame decoder --(crc fail)--> counted separately:
|                                     a transport fault is not a sensor fault
-> parser                          --(reject)----> logged with the reason
-> validator                       team, sequence, duplicates, GPS plausibility
-> link health                      rate, loss %, staleness
-> raw .tsv + parsed .csv + dashboard + web console
```

The receive pipeline runs on a background thread and hands the UI a snapshot through a bounded queue, so a slow interface can never stall reception or logging.

Nothing received is ever discarded. Malformed packets and CRC failures all reach the raw log with their receipt time and the reason they were rejected.

The bridge firmware is a pure bridge: it frames every received payload onto USB with a CRC and emits a status line at 1 Hz carrying its own frame and drop counters. On the first closed link those counters and the packet stream **agreed exactly** — frames incremented once per packet with `dropped=0` — which is how it is known that the radio delivered everything it heard and USB carried everything the radio delivered.

14. Verification and testing

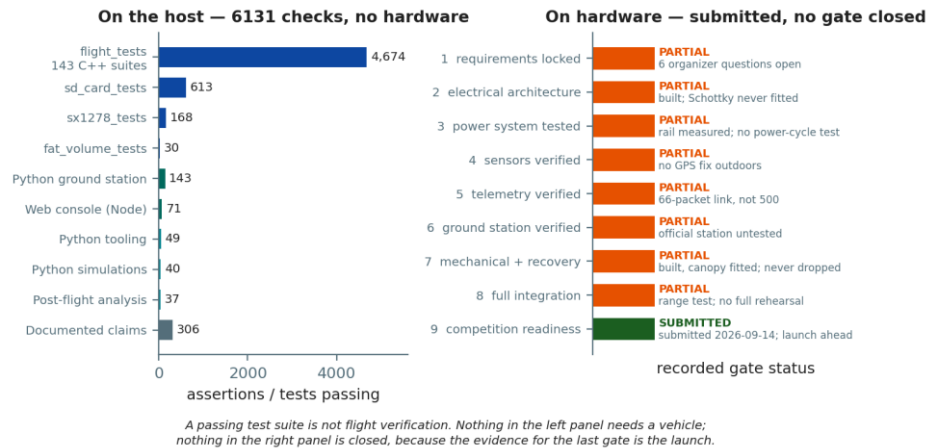


Figure 8 — what is verified on the host, and what is verified on hardware.

```
bash tools/build_host.sh
```

Compiles and runs every C++ suite, the Python suites and the Node suite, then checks the documentation against the source. **All of it runs without hardware.**

Suite	Coverage	Result
flight_tests	143 suites: packet format, parser, shared fixtures, state machine, orientation and angle wrapping, GPS validation, sensor math, timing, calibration, faults, scheduler, block log and torn-header recovery, controller behaviour, link profile, LoRa airtime	4674 / 4674
sd_card_tests	microSD init, SDHC vs SDSC addressing, block round trip, bus release, timeouts, write-error paths	613 / 613
sx1278_tests	LoRa register sequence, TX timeout, RX and CRC handling, RSSI conversion	168 / 168
fat_volume_tests	FAT32 log lookup: MBR and superfloppy, contiguity, missing file, a card that stops answering	30 / 30
flight_smoke_test	Boot, first three packets, GPS parse	Passed
ground_station_tests	Framing, CRC detection, resync, known-answer vector	Passed
Python — ground station	Parser, validator, transport, health, logging robustness, bridge status, vehicle-restart recovery, and a cross-language end-to-end trace of real vehicle output	143 / 143
Python — tooling	LoRa airtime model, pinned to published SX127x reference vectors	49 / 49
Python — simulations	Descent model against closed-form limits, ISA density, mass-tolerance argument	40 / 40
Python — post-flight analysis	Recovers a synthetic flight's known descent rate, drag coefficient, spin, drift and lost packet; the notebook executed cell by cell	37 / 37
Node — web console	Framing, parser, validator, link health, bridge status, extracted from <code>index.html</code>	71 / 71
Documented claims	Numbers in the documentation checked against the source that defines them	306 / 306
Pico syntax	11 translation units against SDK stubs	All OK

Total: 6131 automated checks.

14.1 What is actually proven

- The emitted packet matches the rulebook format **byte for byte**.
- The BMP280 compensation reproduces the **datasheet's own reference vector**.

- A boost before arming **cannot** trigger a launch.
- A hovering drone **cannot** be read as a landing.
- Invalid mandatory data suppresses a packet **without consuming its number**.
- `crc16_ccitt("123456789") == 0x29B1` — the standard known-answer vector.
- The vehicle and the bridge are proven to configure **the same radio modem**.
- Three independent parsers in three languages agree on **one fixture file**.

14.2 What is not covered

The mechanical system, a real descent, and any of it in flight. The sensors, the radio link, the card and the power rail are no longer on this list — they have been measured — but **a bench is not a flight, and none of this is evidence that the vehicle flies.**

14.3 The documentation checker

`tools/check_doc_claims.py` is unusual enough to be worth describing: it reads numbers out of the documentation and checks them against the source that defines them — pin assignments, telemetry rates, watchdog periods, packet sizes, test counts, rulebook constants, the generated netlist, the descent model's own output, the mass budget's arithmetic, and every relative link and heading anchor across 53 documents.

It has caught real errors repeatedly, including two during the writing of this report. A documentation set that cannot be checked drifts from the code within days; this one fails the build instead.

15. Bring-up measurements

Every prediction was written down *before* the measurement was taken, and both are recorded side by side in `documentation/testing/bring-up-record.md` — 94 rows, of which 34 have been taken.

15.1 Highlights

Quantity	Predicted	Measured
Radio version register	0x12	0x12
Airtime, 206-byte packet, SF7/125 kHz	328 ms	333.7 ms — +1.8 %, reproduced to 0.1 ms across four sessions
Gyro bias, per axis	Within ±25 dps	X -3.30, Y +0.86, Z -0.06 dps
Barometer output rate	83 Hz	83.0 Hz
GPS NMEA output	Clean sentences	All six, 162 B/s, 0 checksum errors
SD single block write	—	2.678 ms mean, 4.9 ms typical worst, 29.8 ms occasional
SD sustained rate	—	~300 writes/s , 3672 in ten seconds
Rail under 45 transmits	Holds	3.28–3.29 V
Rail at 100 % write duty	Holds	3.28–3.30 V
Sensor read cost	Under 33 ms	0.833 ms worst
Altitude at rest after calibration	≈ 0.0 m	–0.5 m to +0.1 m across 62 packets
First link	—	66 packets, P-001–P-066, no gaps, no duplicates, 0 % loss, –44 dBm
MAX_RATE at the station	~3.13 Hz	3.11 Hz , 1 packet in 544 lost, RSSI –21 dBm, SNR 10.0 dB

15.2 The measurement that characterises the vehicle rather than testing it

Yaw drift, from a 36.8-minute stationary log of 2209 records. Yaw made **more than a full revolution while the vehicle sat still** — unwrapped, it ran -16.9° to $+362.8^\circ$ and back to $+79.0^\circ$.

But it did not drift steadily, and that is the useful part. **The first 15 minutes were nearly perfect: $\pm 0.8^\circ$ total, at 0.002 dps.** Drift then switched on to ± 0.39 dps and later reversed. Something changed at $t \approx 900$ s, and 0.4 dps is what it costs — **70° of yaw error over a 180-second flight.**

Two candidates and the data cannot separate them: board temperature moved from flat to rising at about the same moment, and the GPS acquired its first fix at $t = 1073$ s and lost it at $t = 1812$ s, which brackets the positive-drift phase almost exactly. But dT/dt does not track the sign. **The experiment that separates them is one stationary re-run with the GPS antenna disconnected**, and it has not been done.

16. Findings register

Twenty-one findings are recorded across receiving inspection, bring-up and the post-flight analysis. The ones that changed the design:

#	What was expected	What happened	Outcome
F-1	An MPU-9250: WHO_AM_I 0x71/0x73, AK8963 at 0x0C	0x70 — an MPU-6500. No magnetometer. 0x0C never appears	Open. The vehicle has no absolute yaw. A procurement item, not a code change
F-5	A radio that either works or does not	The same transmit failed 5/5 and then succeeded 45/45 minutes later, same code, same antenna	Closed by F-10 — the RA-02's 3V3 jumper
F-10	A card that reads is a card that writes	With a long 3V3 jumper the card initialised, read its BPB and FAT, then went silent on the very first write, five runs running. A short jumper fixed it outright: 100/100	Closed. The write-current spike arrives down whatever the supply wire can deliver; reads never draw enough to expose it. On the soldered board this became a short track and a 470 μ F bulk capacitor
F-11	Write latency small against the 33 ms sensor period	29.8 ms worst case , seen twice in five sessions on two boards	Open. One telemetry append is <i>two</i> writes, so a bad second can block the loop for ~ 56 ms — close to two sensor periods
F-12	The card to initialise as reliably as it had	3 failures in the first 4 runs, then 11 consecutive passes , with the supply measured innocent	Bounded, not closed. Ten clean runs bound the rate at only $\sim 30\%$ at 95 % confidence. The clustering favours <i>something changed over random</i> — and nobody can say what
F-16	The card to work under the flight image as under the bring-up image	It failed every time under the flight image while passing twelve times under bring-up	Root cause found, and it is not the card. lora_cs was configured only by the radio driver, which runs <i>second</i> — so through the whole SD initialisation GP17 was an unconfigured pad, and an RP2040 pad resets with its pull-down enabled, which holds it LOW , which on the RA-02 means selected . The radio drove MISO for the entire sequence. Fixed by driving both chip selects high with the bus
F-17	Yaw to drift steadily	It did not drift at all for 15 minutes, then drifted hard	Open. 0.39 dps is 70° over a 180 s flight
F-18	A reported fix to be a holdable fix	739 fixes from a stationary receiver: median 8.1 m from the centroid, max 50.9 m, and the largest single-second jump 55.6 m	Gated. The parser rejected only <code>quality <= 0</code> and never tested satellite count at all. It now refuses a fix below 4 satellites or above HDOP 5.0, and parses HDOP — which it previously discarded
F-20	Landing detection to need a landing	A landing declared under a hovering drone, 12 s before release	Closed by the descent gate

#	What was expected	What happened	Outcome
F-21	The vehicle's altitude to be its height above the pad	About 5 % low on a hot day. The firmware's ISA formula assumes 15 °C; real height per pascal scales with the real temperature	Corrected in the analysis , since the sealed image cannot change: the descent rate is taken from temperature-corrected height, which matters because the uncorrected rate reads ~5 % low — enough to make a 5.05 m/s descent look compliant

Two things are worth noticing about that list. Every one of these came from *running* something rather than from reading it. And **the defects a host suite of five thousand assertions could not find — the chip-select ordering, the supply jumper, the wrong IMU — are all of the kind that only appear when a real board is powered up.**

17. Project timeline

The project ran from 3 September to submission on 14 September 2026 — **twelve days, over two hundred commits.**

Days	What happened
09-03 → 04	The whole software stack, written and tested on the host, then a second pass over it that found nine defects in code that already built and passed — a telemetry rate the radio could not have delivered, three parsers that disagreed, sensors that could not feed their own loop, an attitude filter wrong at the $\pm 180^\circ$ seam, a packet budget below the real packet, and two SPI drivers that had never executed
09-04 → 05	Parts arrive and are photographed and identified one board at a time. The IMU is not the part that was ordered. The microSD reader turns out to be a 3.3 V board, which closes the blocker that had held up the power design for days
09-05	Breadboard bring-up. The radio answers and transmits, the card initialises and writes, and two separate intermittents both turn out to be one long supply jumper
09-06	The board is <i>designed</i> rather than assembled: floorplan to scale, coupling analysis, decoupling sized against a failure that actually happened, sixteen gated build steps
09-07	The board is built and gates 3–7 pass on it. The first radio link closes — 66 packets, no gaps. Running the flight image finds a chip-select ordering bug no host test could have
09-08	Telemetry to 1.43 Hz, an authorised erase command that is inert by construction, GPS gated on satellite count and HDOP, and the first mission analysis the project has had — which finds F-20
09-09	The mechanical design arrives. Cansat_D1 modelled and exported, the organizers confirm a 12 cm sided box so it fits, PETG chosen and argued, three stress studies run, the electronics weighed at 151.299 g, print orientation decided
09-10 → 11	The first range test, and the discovery that the organizers' station had barely heard the vehicle — it was on the test sync word. Both Picos move to 0xA5, the packet budget drops to 200 bytes, GPS and sound go on the air, and MAX_RATE is measured at 3.11 Hz
09-12	The structure comes back from the printer and the vehicle becomes a physical object. White PETG, electronics mounted, egg chamber fitted, 280 g on a scale. The mass risk inverts and doubles
09-13 → 14	Finished and submitted. The 80 cm canopy sewn and fitted, the switch and power LED fitted, the vehicle ballasted into the 450–550 g band, and the CanSat and this report submitted (reported by the team, 2026-09-14)

17.1 Development gates

Nine gates were defined at the start. **None is closed, and that is the correct reading at submission:** each gate is closed by evidence, and the evidence for the last of them is a flight. Figure 8 shows where each stands.

Gates 1–6 are substantially done and each is held open by a small, named item. **Gate 7 is built but not verified** — structure, egg chamber and canopy all exist, and nothing about recovery has been tested. **Gate 8** has a bench link, a

range test and a measured max-rate pattern, but no full rehearsal. **Gate 9 is at submission:** the vehicle is handed in and the launch is what is left.

17.2 What is left

Nothing blocks the launch. Four things are worth doing before it:

8. **Weigh the submitted vehicle.** The final mass is not on record, and it decides which modelled descent the flight is compared against.
9. **Write the analysis notebook.** **Done** — analysis/: a notebook and a one-command CLI that write the mandatory graphs, the descent rate and drag coefficient, and a summary, tested against a synthetic flight. Rehearse it once before the launch.
10. **Rehearse the pad sequence once:** power on, command window, MAX_RATE, wait for ST-R11.... A drone lifting inside the window is the one mistake that costs the flight its state detection.
11. **Never connect USB while the battery is in.** The Schottky was not fitted.

18. Scoring assessment

Against the 200-point rulebook. **This is an estimate, and half of it is a judgement call about work nobody has done yet** — sections D and F are scored on appearance and presentation, and section C's descent scoring is explicitly comparative across teams.

Section	Max	Secured at submission	Achievable	Gap
A · Payload safety	25	0	5	Egg test declined — 20 points forgone by choice. Canopy built; deployment scored at the launch
B · Telemetry & communication	25	0	23	Scored on what the organizers' station receives in flight
C · Parachute, descent, stability	25	~5	23	Switch and power LED fitted. Descent, stability and landing scored at the launch
D · Structural & material innovation	30	~12	26	Structure, egg chamber and canopy built; build quality judged from photographs
E · Technical design & analysis	70	~29	58	Perfboard not PCB; sensor integration at its cap; analysis needs flight data
F · Final report	25	~20	24	This report, submitted. Imaging not recorded in the repository
Total	200	~66	~159	

The team elected not to fly an egg payload, on personal grounds. The 20 points for egg integrity are treated as forgone. **This is not a disqualification** — the rulebook's disqualification list is closed and specific, and an absent egg appears on none of it. The egg *chamber* was built anyway, because PAY-002 is a separate requirement, it carries the +7 cm dimensional allowance, and section D scores effective use of permitted volume.

18.1 Where the strength is

Code originality — 9 or 10 of 10. Self-written, no third-party libraries in the flight path, heavily commented, held by 6131 automated checks. This section rewards exactly what this repository is.

Sensor integration — 25 of 25, at the cap. The mandatory set is 15, GPS is the first +5, and the microphone is the second. **A third additional sensor scores nothing in this section** — a point worth planning against, because it is easy to assume otherwise.

18.2 Where the gaps are

PCB design — near zero on a perfboard. The section asks for original PCB design with layout diagrams, minimal external wiring and well-routed traces, with a bonus for custom boards *instead of* generic dev boards. A hand-wired prototype board is the opposite of that on every criterion. **The pin map has been frozen and hardware-verified since gate 5**, so a schematic could be drawn today.

Data analysis — 20 points, ready before the flight. `analysis/flight_analysis.py` and its notebook read the SD log, the ground CSV or raw packets, and in one command write the three mandatory graphs, the descent rate by regression, the implied drag coefficient, acceleration, orientation, spin, pendulum frequency, GPS drift, acoustic level and SD-against-ground radio loss. It is tested against a synthetic flight whose answers are known — rate within 3 %, drag coefficient within 6 %, release within 0.15 s — and it corrects F-21's altitude bias.

18.3 What would lose points

- **Stray transmission during another team's launch. The 2026 revision made this five times harsher: −1 point per 2 packets.** At 1.43 Hz that is 0.71 points per second — **35 seconds of a CanSat accidentally left on costs more than the entire telemetry section is worth** — and after its five-minute window the vehicle is at 3.11 Hz, so about sixteen. The firmware is *required* to transmit on power-up, so **the fitted switch is the only control and the dark power LED the only confirmation.**
- **Wrong sync word.** A mismatch is a silent, total loss of telemetry that looks exactly like a dead radio. This has already happened once, on the range test.
- **Wrong team number or a malformed packet** zeroes the telemetry section regardless of how well the link performs.
- **A drone lifting inside the command window.** The data still flows, but FLIGHT, LANDED and the post-impact window are never declared, which is the state record a judge reads.
- **Exceeding the size or mass limit by more than 10 % is a disqualification.** The vehicle was ballasted into the mass band and designed inside the size limit.

19. Open questions for the organizers

Ordered by consequence. One question from the build is closed and kept here struck through, because how it closed is part of the record.

12. **What constitutes valid yaw data?** The delivered IMU is a six-axis MPU-6500 with no magnetometer, so the vehicle can transmit only a relative, gyro-integrated yaw (YR-G in the log). Is that acceptable? **If an absolute magnetic yaw is required, this is a part the vehicle does not have.**
13. **How is the ≤ 5 m/s descent requirement enforced and scored?**
14. **Are any LoRa parameters prescribed beyond the sync words?**
15. **What scoring thresholds apply where the rulebook rewards higher performance?** The 2026 revision rewards packet rates above 1 Hz and longer stable descents but names no thresholds.
16. **Does declining the egg carry any consequence beyond the 20 points**, given the general rules call it mandatory while the disqualification list does not?
17. **Is the $500 \text{ g} \pm 10 \%$ mass limit a band or a ceiling?** **Moot at submission.** At 280 g assembled the vehicle would have been 105–135 g under the lower edge; it was ballasted into the band, which satisfies either reading.

20. Lessons learned

1. **Owning a component is not integrating it, and a listing is not a datasheet.** Two of the delivered parts were not what they were sold as: the IMU is a six-axis MPU-6500 rather than the nine-axis MPU-9250 ordered, and the microSD listing described a board with a regulator when the delivered one had none. Both were found by **photographing and identifying every board on arrival** rather than by assuming. One of them cost a mandatory telemetry field; the other closed a blocker that had held the power design up for days. The inspection paid for itself twice, in opposite directions.
2. **The defects that matter most are the ones a test suite cannot reach.** Five thousand host assertions did not find the chip-select ordering bug, the supply jumper, or the wrong part number. All three needed a powered board. **Write the tests — they caught nine real defects in code that already built — but do not mistake them for evidence about hardware.**
3. **A marginal wire is indistinguishable from a broken component.** Two separate intermittents — a radio that transmitted 5/5 then 0/5, and a card that read perfectly and refused to write — turned out to be **the same fault on the same kind of wire on two different modules**. A supply jumper that carries an idle part will not carry a working one, and reads never draw enough current to expose it. Shortening the jumper fixed both outright.
4. **Estimates are wrong in both directions, and the scale settles it in one minute.** The avionics estimate was 31 g *low*; the structure estimate was 64 g *high*. Both had reasoning behind them and both were labelled as estimates. **Neither survived contact with a scale**, and the second one inverted the project's largest risk twice in three days.
5. **Write the operational document before you think you need it.** The concept of operations was written as documentation, and writing it found F-20 — a vehicle that declared a landing while hanging under the drone, twelve seconds before release. **No test found it because no test described a hover.** Describing the mission in order, minute by minute, was the thing that exposed a state the design had never considered.
6. **Two systems that must agree should have one definition.** The vehicle and the ground bridge carried separate copies of the modem parameters and agreed only by coincidence; three parsers in three languages disagreed about the packet format. Both were fixed the same way — **one shared header, one shared fixture file** — and both would otherwise have failed silently and totally.
7. **Make the documentation fail the build.** `check_doc_claims.py` checks 300 numbers in the documentation against the source that defines them. It has caught real drift repeatedly, including twice while this report was being written. **A documentation set that cannot be checked is stale within a week**, and stale documentation is worse than none because it is believed.
8. **Say what you have not done.** Every status in this project is written against the rule that nothing is claimed without evidence, and the most valuable part of that discipline is not the accuracy — it is that **the list of what is missing stays short and visible**, which is what makes it actionable. The three hardware blockers were closed because they were written down as blockers rather than described as risks.
9. **The test you skip is the one the launch runs for you.** Everything on this vehicle that could be measured on a bench was measured, and most of it twice. The drop test was the one mechanical test that needed a site, a height and an afternoon, and it is the one that was not done before submission — so the canopy's first opening, the structure's first arrival and the model's first check will all happen at the launch, at once, with points on them. The bring-up record shows what happens to a prediction that meets reality: the airtime model survived to within 1.8 %, and the mass estimates did not survive at all.
10. **Re-read the code before repeating a number about it.** The documentation said for a week that the SD log runs at the 30 Hz sensor rate, and this report repeated it. It does not: the row is appended once per telemetry packet. The 30 Hz figure was true of the acquisition loop and quietly became a claim about the log. A stationary bench log of 2209 rows in 36.8 minutes had shown the real rate all along, to anyone who divided.

Appendix A — pin assignment

BoardPins in `firmware/flight-computer/include/flight/config.hpp` is the source of truth. The netlist and the wiring diagrams are **generated from it**, and the generator refuses to run if they disagree.

GPIO	Function	Device
GP4 / GP5	I2C0 SDA / SCL	MPU-6500 (0x68) + BMP280 (0x76)
GP16 / GP18 / GP19	SPI0 MISO / SCK / MOSI	RA-02 + microSD
GP17	Chip select	RA-02
GP6	Chip select	microSD
GP20 / GP21 / GP22	RESET / DIO0 / DIO1	RA-02
GP12 / GP13	UART0 TX / RX	NEO-6M
GP7	Interrupt	MPU-6500 — wired, firmware does not enable it
GP14	Status LED	External LED, blink rate encodes mission state
GP15	Comparator input	LM393 D0
GP26	ADC0	Battery sense, divider ratio 2.0 (33 k Ω / 33 k Ω)
GP27	ADC1	LM393 A0

Appendix B — flight configuration constants

From `firmware/flight-computer/src/pico/main.cpp` and `config.hpp`. Every value is validated at startup by `validate_config()`, which refuses to run a configuration that cannot meet the rulebook.

Constant	Value	Note
<code>team_id</code>	CAN-Team-25	The formatter refuses the rulebook's CAN-Team-XX placeholder
<code>loop_tick_ms</code>	2 ms	Bounded twice: scheduling jitter, and draining the GPS UART before its FIFO fills
<code>sensor_period_ms</code>	33 ms	30 Hz attitude and altitude-rate update
<code>telemetry_period_ms</code>	700 ms	1.43 Hz. Hard ceiling 950 ms, enforced three ways
<code>sd_flush_period_ms</code>	2000 ms	Appends happen per packet; this is the sync
Watchdog	2000 ms	A hung loop reboots; telemetry restarts automatically
<code>calib_samples</code>	80	~2.7 s at 30 Hz
<code>calib_timeout_ms</code>	20 000 ms	Then resolve best-effort
<code>arming_delay_ms</code>	3000 ms	And calibration must have settled
<code>launch_confirm_ms</code>	300 ms	Launch condition must hold this long
<code>launch_altitude_gain_m</code>	15 m	Or acceleration above 30 m/s ²
<code>min_flight_ms</code>	3000 ms	Landing detection suppressed until this into flight
<code>landing_descent_rate_mps</code>	-2 m/s	The descent gate — F-20
<code>landing_descent_confirm_ms</code>	1000 ms	Held this long before the gate opens
<code>landing_confirm_ms</code>	3000 ms	At-rest condition must hold this long
<code>post_impact_transmission_ms</code>	5000 ms	≥ the rulebook's 5 s ; a shorter value will not build
<code>sensor_stale_after_ms</code>	2000 ms	Time-based, not attempt-based

Constant	Value	Note
sd_max_failures	10	Consecutive write failures before logging disables itself
radio_recovery_backoff_ms	1000 ms	Bounded re-init, never a blocking retry
battery_divider_ratio	2.0	33 kΩ / 33 kΩ into GP26
battery_low_voltage	3.5 V	Raises a warning; nothing more
command_window_ms	300 000 ms	Five minutes from power-on, pre-arm only
auto_max_rate	true	Goes to max rate when the window closes, and after a watchdog reset

All detection thresholds are marked **PROVISIONAL** and want tuning against real drop data.

Appendix C — repository map

firmware/	
common/	shared telemetry format + SX1278 driver (cansat::)
flight-computer/	flight core + Pico HAL + tests (flight::)
ground-station/	bridge firmware + USB CRC framing (ground::)
ground-station/	
software/	Python receive pipeline + tests
web/	single-file browser telemetry console
avionics/	per-subsystem summaries against what was measured
electrical/	
schematics/	machine-readable netlist, generated from the firmware
PCB/	board layout – perfboard today, nothing fabricated
mechanical/	envelope, mass budget, canopy spec, CAD, stress studies
simulations/	descent model + tests
analysis/	post-flight analysis: notebook, one-command CLI, synthetic flight
tools/	host build, Pico syntax check, LoRa link-budget calculator, netlist and drawing generators, documentation-claim checker, SD-card and flight-log utilities, report figures, SDK stubs
test-data/	shared fixtures the C++, Python and JavaScript parsers all read
documentation/	
requirements/	rulebook, requirement checklist, gates
mission/	concept of operations
design/	architecture, protocol, wiring, electrical, link budget
hardware/	BOM, inspection, assembly, compatibility, GPIO map, datasheets
project/	timeline, scoring, this report
testing/	test plan and the bring-up measurement record
operations/	runbook and launch-day procedure
audit/	repository audits

Submitted 14 September 2026. Nothing in this project is claimed as flown.

The software is built and tested. The board is built, and every device on it has answered on a bench. The structure is printed, the canopy and the switch are fitted, and the vehicle is in the mass band.

The launch is still ahead, and it will be the first time the vehicle descends under its canopy.